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CSE 160 (Fall 2020) – Homework 1

Due: Oct 12 at 11:59pm PT on Canvas.
Total: 10 points.

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1. Normalize the vector v = [1,2,3]. Round your answer to 2 decimal points.

   Solution.

$$\hat{v} = \frac{\vec{v}}{\|v\|}$$

$$\|v\| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14} = 3.7416 \approx 3.74$$

$$\hat{v} = \frac{[1,2,3]}{3.74} = [0.27, 0.53, 0.80]$$

2. Simplify and Find the Length of the Vector

   (a) $3 * [1,1] + 2 * [-1,1]$

   Solution.

$$3 * [1,1] + 2 * [-1,1]$$
$$= [3,3] + [-2,2]$$
$$= [1,5]$$

   (b) Find the length of the vector valculated in (a)

   Solution.

$$\textit{Length of vector: } \|[1,5]\|$$

$$\|[1,5]\| = \sqrt{1^2 + 5^2} = \sqrt{26} = 5.099 \approx 5.10$$

3. Calculate the Cross Product

   (a) $[0,1,1] \times [1,1,0]$

   Solution.

$$\begin{bmatrix} i & j & k \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$= i[1 \cdot 0 - 1 \cdot 1] - j[0 \cdot 0 - 1 \cdot 1] + k[0 \cdot 1 - 1 \cdot 1]$$

$$= i[-1] - j[-1] + k[-1] = [-1, 1, -1]$$

   (b) $[2,3,4] \times [1,0,0]$

Solution.

$$\begin{bmatrix} i & j & k \\ 2 & 3 & 4 \\ 1 & 0 & 0 \end{bmatrix}$$

$$= i[3 \cdot 0 - 4 \cdot 0] - j[2 \cdot 0 - 4 \cdot 1] + k[2 \cdot 0 - 3 \cdot 1]$$

$$= i[0] - j[-4] + k[-3] = [0, 4, -3]$$

(c) $[0, 3, 4] \times [2, 2, 2]$

Solution.

$$\begin{bmatrix} i & j & k \\ 0 & 3 & 4 \\ 2 & 2 & 2 \end{bmatrix}$$

$$= i[3 \cdot 2 - 4 \cdot 2] - j[0 \cdot 2 - 4 \cdot 2] + k[0 \cdot 2 - 3 \cdot 2]$$

$$= i[-2] - j[-8] + k[-6] = [-2, 8, -6]$$

4. Calculate the Dot Product

(a) $[1, 0, 1] \cdot [0, 1, 1]$

Solution.

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = (1 \cdot 0) + (0 \cdot 1) + (1 \cdot 1) = 1$$

(b) $[0, 3, 4] \cdot [1, 0, 0]$

Solution.

$$\begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = (0 \cdot 1) + (3 \cdot 0) + (4 \cdot 0) = 0$$

(c) $[2, 3, 4] \cdot [6, 4, 3]$.

Solution.

$$\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 6 \\ 4 \\ 3 \end{bmatrix} = (2 \cdot 6) + (3 \cdot 4) + (4 \cdot 3) = 12 + 12 + 12 = 36$$

5. Consider a triangle formed by connecting the three points p_1 = (0,0,0), p_2 = (1,0,0) and p_3 = (1,1,1).

(a) Find the area of the surface of this triangle.

Solution.

$$\textit{Area of a triangle given 3 points: } \frac{1}{2} \cdot \|\overrightarrow{p_1p_2} \times \overrightarrow{p_1p_3}\|$$

$$\overrightarrow{p_1p_2} = (0, 0, 0) - (1, 0, 0) = [-1, 0, 0]$$

$$\overrightarrow{p_1p_2} = (0, 0, 0) - (1, 1, 1) = [-1, -1, -1]$$

$$\begin{bmatrix} i & j & k \\ -1 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

$$= i[0 \cdot -1 - 0 \cdot -1] - j[-1 \cdot -1 - 0 \cdot -1] + k[-1 \cdot -1 - 0 \cdot -1]$$

$$= i[0] - j[1] + k[1] = [0, -1, 1]$$

$$\|[0, 1, 1]\| = \sqrt{0^2 + 1^2 + 1^2} = \sqrt{2}$$

$$\textit{Area of triangle: } \frac{\sqrt{2}}{2}$$

(b) Find the vector, which is perpendicular to the surface of this triangle, AND has a positive z-direction.

Solution. *Perpendicular vector to the surface of the triangle is the cross product of 2 vectors defining the sides of the triangle and the result is as seen in part (a):*

$$[0, -1, 1]$$

6. Calculate the Matrix

(a) $\begin{bmatrix} 1 & 2 & 5 \\ -1 & -1 & 1 \\ 4 & 4 & -2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

Solution.

$$\begin{bmatrix} 1 & 2 & 5 \\ -1 & -1 & 1 \\ 4 & 4 & -2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} (1 \cdot 1) + (2 \cdot 2) + (5 \cdot 3) \\ (-1 \cdot 1) + (-1 \cdot 2) + (1 \cdot 3) \\ (4 \cdot 1) + (4 \cdot 2) + (-2 \cdot 3) \end{bmatrix} = \begin{bmatrix} 1 + 4 + 15 \\ -1 + -2 + 3 \\ 4 + 8 + -6 \end{bmatrix}$$

$$= \begin{bmatrix} 20 \\ 0 \\ 6 \end{bmatrix}$$

(b) $\begin{bmatrix} 2 & 0 & 3 \\ 0 & -1 & 2 \\ 4 & 2 & -2 \end{bmatrix} \cdot \begin{bmatrix} -1 & -4 & 1 \\ 1 & -1 & 4 \\ 0 & 0 & 5 \end{bmatrix}$

Solution.

$$\begin{bmatrix} 2 & 0 & 3 \\ 0 & -1 & 2 \\ 3 & 2 & -2 \end{bmatrix} \cdot \begin{bmatrix} -1 & -4 & 1 \\ 1 & -1 & 4 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} (-1 \cdot 2) + (1 \cdot 0) + (0 \cdot 3) & (-4 \cdot 2) + (-1 \cdot 0) + (0 \cdot 3) & (1 \cdot 2) + (4 \cdot 0) + (5 \cdot 3) \\ (-1 \cdot 0) + (1 \cdot -1) + (0 \cdot 2) & (-4 \cdot 0) + (-1 \cdot -1) + (0 \cdot 2) & (1 \cdot 0) + (4 \cdot -1) + (5 \cdot 2) \\ (-1 \cdot 3) + (1 \cdot 2) + (0 \cdot -2) & (-4 \cdot 3) + (-1 \cdot 2) + (0 \cdot -2) & (1 \cdot 3) + (4 \cdot 2) + (5 \cdot -2) \end{bmatrix}$$

$$= \begin{bmatrix} -2 + 0 + 0 & -8 + 0 + 0 & 2 + 0 + 15 \\ 0 + -1 + 0 & 0 + 1 + 0 & 0 + -4 + 10 \\ -3 + 2 + 0 & -12 + -2 + 0 & 3 + 8 + -10 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & -8 & 17 \\ -1 & 1 & 6 \\ -1 & -14 & 1 \end{bmatrix}$$

7. Consider two lines $y = \frac{4}{3}x - 1 and y = 0$:

   (a) Find the intersection point between the two lines (Draw the lines on a graph if stuck).

   Solution.

$$0 = \frac{4}{3}x - 1$$
$$1 = \frac{4}{3}x$$
$$\frac{3}{4} = x$$
$$\textit{Intersection Pt: } (\frac{3}{4}, 0)$$

   (b) Are these lines perpendicular?

   Solution. *Perpendicularity is defined as the slopes of 2 intersecting lines being negative reciprocals of each other. In this case the slopes are* $\frac{4}{3}$ *and* 0 *which are clearly not negative reciprocals of each other.*

Readings Ch1 & Ch2

1. Define OpenGL, OpenGL ES and WebGL. Describe their relationship.

   Solution. *OpenGL is the dedicated computer library akin to Direct3D. Ususally interfaced with low level programming languages, OpenGl is a graphics library to render elements on a desktop system. To use the graphics library for less powerful devices, the library got a variant at version 2.0 called OpenGL ES. The ES stands for Embedded Systems. Similarly at version 2.0 of OpenGL WebGL was also developed as OpenGL for the browser. WebGl is built into the browser enableing the 3D graphics library to interface with runtime scripted languages such as HTML and Javascript whose fundamental design revolves around instantaneous update just by changing the file contents (C/C++ requires the developers to compile the code to run on the end user's devices).*

2. What is the difference in the software architecture of a webpage and a webpage using WebGL. Describe all components involved in the two cases.

   Solution. *A Web page without WebGL only has HTML, JavaScript, and the HTML Rendering Engine as components of the browser running the page. A Web page with WebGL on the otherhand includes the components GLSL ES, WebGL, OpenGL/OpenGL ES as well. GLSL is the OpenGL shading language and WebGL runs on top of the OpenGL/OpenGL ES runtime.*

3. In HTML, what is the <canvas> tag?

   Solution. *The <canvas> tag was an addition to the HTML5 standard that offered a less complicated way for computer graphics to be prgramatically generated through the Javascript language. The <canvas> can draw almost anything in the area as programmed.*

4. In Javascript, what line of code one writes to retrieve a 2D rendering context?

Solution.

```
// 2D context
var ctx = canvas.getContext("2d");
```